

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0407

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
 2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
 3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
 4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
 5. Given: Input (x) = 4, Weight (w) = 2, Actual Output = 10, Prediction: $y = w \times x$
 - 1) Find the predicted output.
 - 2) Calculate the error.
 - 3) If the gradient is 4 and learning rate is 0.01, find the updated weight.
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Code of Conduct:

*I **P. DEEKSHANA / 192425104** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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CO1 – AT1 – ANALYTICAL PROBLEM SOLVING

NAME: P. DEEKSHANA

REG NO: 192425104

COURSE CODE: MLA0407

COURSE NAME: DEEP LEARNING

Q1. Maximum Likelihood Estimation (MLE) for Coin Toss

Given: Number of tosses, $n = 10$

Outcomes:

H H T H H H T H H T

Number of Heads = 7

Number of Tails = 3

Formula:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation:

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer: MLE estimate of the probability of getting Heads is

$$\boxed{\hat{p} = 0.7}$$

Interpretation: The estimated probability of getting a Head is 70%.

Q2. MLE for Student Pass Probability

Given: Total students = 50

Students passed = 42

Formula:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Calculation:

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer:

$$\hat{p} = 0.84$$

Interpretation: The estimated probability that a student passes is 84%.

Q3. MLE for Defective Light Bulbs

Given: Total bulbs = 100

Defective bulbs = 8

Formula:

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation:

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer:

$$\hat{p} = 0.08$$

Interpretation: The estimated probability of a bulb being defective is 8%.

Q4. MLE for Rolling a 6

Given: Total rolls = 60

Number of times 6 appears = 18

Formula:

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation:

$$\hat{p} = \frac{18}{60} = 0.30$$

Answer:

$$\hat{p} = 0.30$$

Interpretation: The estimated probability of rolling a 6 is 30%.

Q5. Neural Network Weight Update

Given:

Input, $x = 4$

Weight, $w = 2$

Actual Output = 10

Prediction Formula:

$$y = w \times x$$

Gradient = 4 , Learning Rate = 0.01

1) Find the Predicted Output:-

Formula:

$$y = w \times x$$

Calculation

$$y = 2 \times 4 = 8$$

Answer

$$\text{Predicted Output} = 8$$

2) Calculate the Error:-

Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation:

$$\text{Error} = 10 - 8 = 2$$

Answer

$$\boxed{\text{Error} = 2}$$

3) Update the Weight:-

Gradient Descent Formula:

$$w_{\text{new}} = w - \eta \times \text{gradient}$$

where

- Learning Rate (η) = 0.01
- Gradient = 4

Calculation:

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

Answer:

$$\boxed{w_{\text{new}} = 1.96}$$

Interpretation: The weight is reduced from 2.00 to 1.96, helping the model reduce prediction error during training.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____